



BACHELOR OF INFORMATION TECHNOLOGY (HONS)

**FINAL EXAMINATION
JUNE 2024**

Course : EC3114 (Software Modelling and Analysis) Time: 2.00 pm – 5.00 pm (3 hours)
Lecturer : Mr. Izrulfizal Saufihamizal bin Ibrahim Date : 1 October 2024

Instructions:

Answer **ALL** questions in the Answer Booklet provided.

The maximum number of marks is 100.

This marking scheme consists of 2 printed pages.
(excluding front cover)

DO NOT OPEN THIS BOOKLET UNTIL YOU ARE TOLD TO DO SO

1. a. Analyse the techniques used for requirements elicitation. Which technique would be most effective for a system that interacts with a wide range of users? Justify your answer. (8 MARKS)
- b. Discuss the differences between functional requirements, non-functional requirements, and constraints. Provide examples of each from a hospital management system. (5 MARKS)
- c. Discuss the deliverables of each phase of the Software Development Life Cycle (SDLC). Which phase do you consider most critical for success and why? (12 MARKS)
2. a. Explain how Use Case Diagrams support functional requirement analysis. (5 MARKS)
- b. Define the concept of a Data Dictionary and explain its importance in system design. (5 MARKS)
- c. Explain the concept of Object-Oriented Analysis and Design (OOAD). Discuss how OOAD principles can be applied to develop a robust software architecture. (10 MARKS)
- d. Discuss the role of screen prototypes in validating system design with stakeholders. (5 MARKS)
3. You are tasked with designing a class diagram for a simple online shopping system. The system should include functionalities for managing products, customers, and orders. Your class diagram should include at least three classes, with attributes, methods, and relationships clearly defined. Provide explanations for your design choices and any assumptions you make. (25 MARKS)

4. Create an activity diagram to model the process of online food ordering on a restaurant platform. The process should include activities such as browsing the menu, selecting items, adding them to the cart, placing an order, payment processing, and order confirmation. Clearly label all activities, decision points, and control flows. Additionally, explain how this activity diagram can help in optimizing the platform's efficiency and user satisfaction.

(25 MARKS)

-END OF QUESTION PAPER-